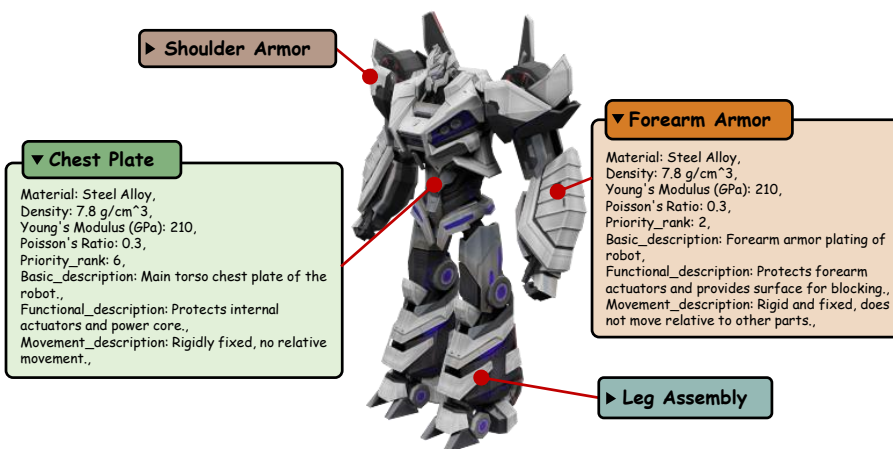
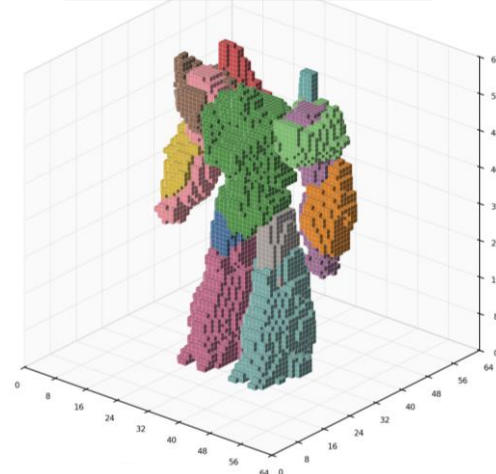


Sim-Ready Physical 3D Asset

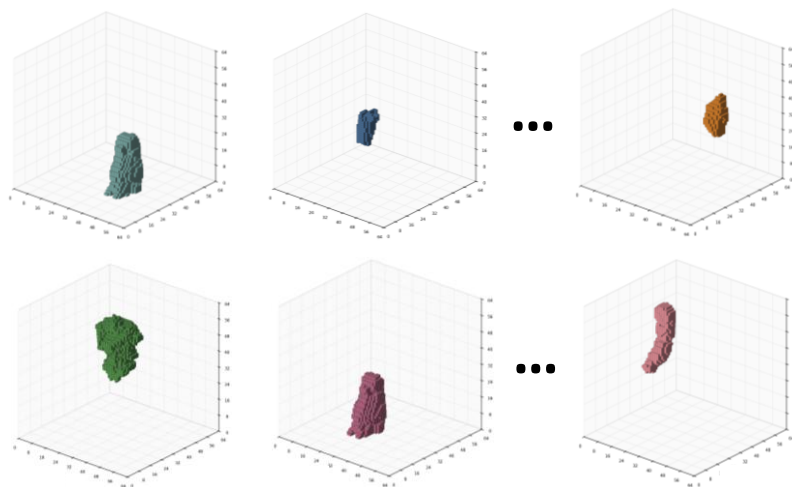


Voxelize

Full Voxel



Part-Level Voxel



2D RLE

Z=0 : Empty

⋮

Z=9 : Empty

Z=10: [2013, 6; 2072, 13; 2172, 9; ... ; 2672, 3]

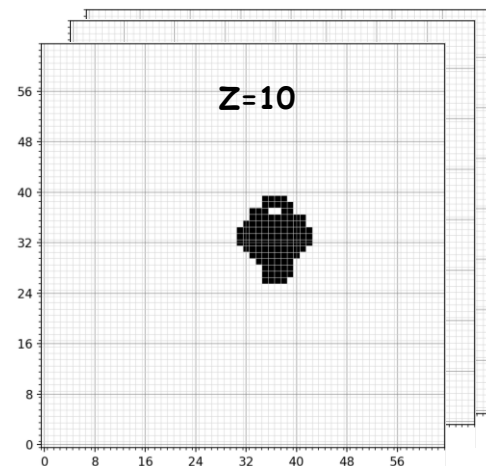
Z=11: [2013, 6; 2072, 13; 2172, 4; ... ; 3142, 8]

⋮

Z=63: [2031, 4; 2056, 7; 2410, 11; ... ; 3962, 5]

Blue: Start voxel index Brown: Length

2D Mask



Z-axis
slicing

Template-based 2D RLE

Template_a: Empty

⋮

Template_e: [2013, 6; 2072, 13; 2172, 9]

Z=0 : layer a

⋮

Z=9 : layer a

Z=10: layer e + ... ; [2672, 3]

Z=11: layer e - [2176, 5; ... ; 3142, 8]

⋮

Z=63: [2031, 4; 2056, 7; 2410, 11; ... ; 3962, 5]